



SILVER OAK UNIVERSITY
Silver Oak College of Computer Application
Master of Computer Application
Course Name: Foundation of Mathematics
Course Code: MDC5001C
Semester: 1st

Prerequisite:

1. Basic algebraic operations, equations and mathematical reasoning.
2. Fundamental concepts of set theory, relations, and functions.
3. Elementary calculus including differentiation and integration.
4. Matrix operations, determinants, and system of linear equations.
5. Introductory discrete structures such as logic, graphs, and trees.

Course Objectives:

1. To develop a strong foundation in combinatorics and counting techniques applicable to computer science problems.
2. To introduce numerical methods for solving algebraic, transcendental, and linear system equations with error analysis.
3. To equip students with Operations Research tools — Linear Programming, Transportation, Assignment and Game Theory for decision-making.
4. To explore algebraic structures including lattices and Boolean algebra, and understand their applications in digital logic and computer science.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
4	60	0	0	-	-	20	40	120	4

Contents:

Unit	Topics	Teaching Hours	% Weightage
1	Introduction to Combinatorics and Its Applications: Basic Counting Principles – Addition and Multiplication Rules; Permutations (With and Without Repetition); Combinations and Related Identities; Binomial Theorem and Pascal's Triangle; Pigeonhole Principle (Basic and Generalized); Inclusion-Exclusion Principle; Derangements and Applications; Counting with Repetition and Multisets; Integer Partitions and Compositions; Recurrence Relations and Their Solutions; Generating Functions and Applications	12	20%
2	Numerical Methods: Types of Errors: Absolute, Relative, Percentage; Solutions of Algebraic and Transcendental Equations – Bisection Method, Regula-Falsi Method, Newton-Raphson Method, Secant Method; Solution of Linear System of Equations – Gauss Elimination, Gauss-Jordan, Gauss-Seidel and Jacobi Iterative Methods; Interpolation – Newton's Forward	20	34%

	and Backward Interpolation; Lagrange's Interpolation; Newton's Divided Difference Interpolation		
3	Operations Research and Linear Programming: Basic concepts, scope and applications of OR; Graphical solution for two-variable LPP; Feasible region, optimal solution, multiple/unbounded/infeasible cases; Standard form of LPP; Introduction to Simplex algorithm and Tableau method; Concept of Duality, formulation of Dual Problem Transportation Problem (TP) – IBFS using North-West Corner Rule and Least Cost Method; Optimality test using MODI method; Degeneracy in TP; Assignment Problem (AP) – Hungarian Method; Introduction to Game Theory: Two-person zero-sum games, saddle point, pure and mixed strategies, dominance principle, graphical method; Real-life applications.	15	24%
4	Lattices and Boolean Algebra: Relations and ordering, Partially Ordered Sets (POSETs); Lattices as POSETs – properties, sub-lattices, direct product, homomorphism; Complete lattices, bounds, distributive lattice, complemented lattices; Introduction and properties of Boolean Algebra; Sub-Boolean algebra, join-irreducible, meet-irreducible, atoms, anti-atoms; Stone's Representation Theorem (without proof); Applications	13	22%
Total		60 Hours	100%

Course Outcomes:

CO No.	CO Statement	Blooms Taxonomy Level	Unit
CO-1	Apply combinatorial techniques — including counting principles, permutations, combinations, binomial identities, recurrence relations, and generating functions — to solve discrete mathematical problems in computer science.	Apply	1
CO-2	Analyze numerical errors and implement root-finding methods, linear system solvers, and interpolation techniques to solve mathematical problems effectively.	Analyze	2
CO-3	Formulate and solve problems in Linear Programming, Transportation, Assignment, and Game Theory using graphical, simplex, Hungarian, and other standard methods, and apply them to real-life decision-making.	Evaluate	3
CO-4	Analyze the structure and properties of lattices and Boolean algebras, and apply these algebraic concepts to understand logical systems, digital circuits, and their applications in computer science.	Apply	4

Teaching & Learning Pedagogy approaches:

Approach	Method	Tools / Platform
ICT-Enabled Teaching	Slide-based lectures	PowerPoint, GeoGebra, MATLAB Online

Experiential Learning	Hands-on programming labs	NumPy, Desmos
Case Studies	Industry optimization scenarios	Excel Solver, OR-Tools
Problem-Based Learning	Weekly problem sets	Assignment sheets, Google classroom
Flipped Classroom	Pre-recorded lectures for self-study	NPTEL, Swayam, YouTube
Project-Based Learning	Term work group projects	LaTeX reports, Google Colab

List of Tutorials: NA

Sr. No.	Tutorial Content	Hours	Co No.
1			
2			
3			

List of Experiments: NA

Sr. No.	Practical Name	Hours	Co No.

Major Equipment and Simulation Tools:

1. Scientific Calculator
2. Computer / Laptop with NumPy, Desmos
3. Microsoft Excel with Solver Add-in
4. MATLAB / GNU Octave (for numerical computations)
5. GeoGebra (for graphical LPP solutions)
6. LaTeX reports, Google Colab (for term work)

Books Recommended(APA Format):

1. Cameron, P. J. *Combinatorics: Topics, techniques, algorithms*. Cambridge University Press.
2. Chapra, S. C., & Canale, R. P. *Numerical methods for engineers* (7th ed.). McGraw-Hill Education.
3. Taha, H. A. *Operations research: An introduction* (10th ed.). Pearson.
4. Bazaraa, M. S., Jarvis, J. J., & Sherali, H. D. *Linear programming and network flows* (4th ed.). Wiley.
5. Kolman, B., Busby, R. C., & Ross, S. C. *Discrete mathematical structures* (6th ed.). Pearson.

Term Work Module:**20 HOURS**

Objective: To reinforce theoretical knowledge through hands-on problem solving, mini-projects, and case studies aligned with real-world applications.

Term Work	Statement	Hours	CO No.
Term Work 1	Mini Project: Identify a real-world combinatorics problem (e.g., scheduling, coding theory) and solve it using counting principles, recurrence relations, or generating functions. Submit a written report with problem formulation, solution, and code (if any).	4 Hours	CO-1
Term Work 2	Numerical Methods Case Study: Select an engineering or science problem and solve it using two numerical methods (root-finding + interpolation). Compare solutions, discuss error analysis, and present computational results using tools like Python/MATLAB/Excel.	6 Hours	CO-2
Term Work 3	Operations Research Problem Solving: Formulate and solve a real-life optimization problem using LPP (Simplex), Transportation (MODI), Assignment (Hungarian), or Game Theory. Present the formulation, solution steps, and sensitivity analysis. Group of 2–3 students.	6 Hours	CO-3
Term Work 4	Lattice and Boolean Algebra Assignment: Design a digital logic circuit for a given Boolean expression, minimize it using Boolean algebra laws, and represent the corresponding lattice structure. Verify with a truth table.	4 Hours	CO-4
Total Hours		20	

Self Learning Module:**40 HOURS**

Objective: To encourage independent, self-paced learning using curated online resources, MOOC platforms, and supplementary materials.

Recommended Online Resources:

Sr. No.	Unit Name / Topic Name	Resource Link
1	Combinatorics – Permutations and Combinations	https://www.khanacademy.org/math/statistics-probability/counting-permutations-and-combinations
2	Generating Functions and Recurrence Relations (NPTEL)	https://nptel.ac.in – Discrete Mathematics course
3	Numerical Methods – Root Finding and Interpolation	https://nptel.ac.in – Numerical Methods course
4	MIT Open Course Ware – Numerical Methods	https://ocw.mit.edu/courses/18-330
5	Operations Research – LPP and Simplex Method	https://swayam.gov.in – Operations Research MOOC

6	Transportation and Assignment Problems (YouTube)	https://www.youtube.com – NPTEL OR Playlist
---	--	---

CO-PO-PSO Matrix:

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1	3	3	2	2	1	2		3	2
CO-2	3	2	3	2	2	2		2	3
CO-3	3	3	2	3	2	2		2	3
CO-4	3	2	2	2	1	2		3	2

CO-PO-PSO MAPPING JUSTIFICATION FOR Foundation of Mathematics:

CO No.	CO Statement	PO No.	Justification
CO1	Apply combinatorial techniques including counting principles, permutations combinations, binomial identities, recurrence relations, and generating functions to solve discrete mathematical problems in computer science.	PO1	Combinatorial mathematics forms an important foundation for computer science and algorithmic problem solving. Concepts such as permutations, combinations, recurrence relations, and generating functions are widely used in programming, data structures, probability analysis, coding theory, and algorithm design. Students develop mathematical reasoning and abstraction skills required for computational modeling and efficient problem-solving.
		PO2	Students analyze complex counting and arrangement problems using combinatorial techniques and logical reasoning. They learn to formulate mathematical models for discrete problems and apply systematic approaches to derive optimal solutions. This strengthens analytical thinking and problem interpretation abilities required in computing disciplines.
		PO3	Students apply combinatorial concepts for developing structured and efficient problem-solving methods in areas such as network design, cryptography, algorithm optimization, and logical computation. These mathematical techniques support the development of computational solutions and algorithmic frameworks.
		PO4	Students investigate recurrence-based structures, pattern generation, and combinatorial relationships through mathematical analysis and logical interpretation. They perform problem exploration and derive conclusions using systematic mathematical investigation methods.
		PO5	Students use digital learning resources, computational software, graphing tools, and AI-assisted platforms for understanding combinatorial

			structures and recurrence patterns. Technology-supported activities improve visualization and mathematical problem-solving skills.
		PO6	Combinatorial problem-solving develops logical decision-making, accuracy, and structured reasoning that are essential in professional computing practices and responsible computational analysis.
		PSO1	Combinatorial methods are directly connected with algorithm design, recursive programming, optimization techniques, and computational logic used in software development and modern programming environments.
		PSO2	Students apply combinatorial thinking in real-world computational applications such as scheduling systems, cryptographic methods, artificial intelligence models, and network optimization problems. These activities encourage innovative and technology-oriented problem-solving abilities.
CO2	Analyze numerical errors and implement root-finding methods, linear system solvers, and interpolation techniques to solve mathematical problems effectively	PO1	Numerical methods provide mathematical foundations for scientific computing, approximation techniques, and computational analysis. Students gain knowledge of error estimation, root-finding algorithms, interpolation, and matrix methods that are essential in computer-based mathematical problem solving.
		PO2	Students analyze numerical accuracy, convergence behavior, approximation errors, and computational efficiency while solving mathematical and engineering problems. These activities strengthen analytical and interpretational abilities.
		PO3	Students implement numerical algorithms such as Newton-Raphson method, Gauss elimination, and interpolation techniques to develop computational solutions for practical mathematical problems.
		PO4	Students investigate the behavior of numerical methods, compare accuracy levels, analyze iterative processes, and interpret computational results using mathematical reasoning and experimentation.
		PO5	Students use numerical computing tools, spreadsheets, programming environments, and mathematical software for solving equations, performing interpolation, and analyzing numerical data efficiently.
		PO6	Accurate numerical analysis and interpretation are important for responsible decision-making in scientific, industrial, and technological applications where precision and reliability are essential.
		PSO1	Implementation of numerical algorithms improves programming logic, computational efficiency, and software-based mathematical modeling skills required in modern computing applications.

		PSO2	Numerical methods are widely applied in simulation systems, machine learning, data analytics, scientific computing, and modern technological applications requiring computational approximation and predictive analysis.
CO3	Formulate and solve problems in Linear Programming, Transportation, Assignment, and Game Theory using graphical, simplex, Hungarian, and other standard methods, and apply them to real-life decision-making.	PO1	Operations Research techniques provide mathematical and computational foundations for optimization, resource allocation, and strategic decision-making in computing and business applications.
		PO2	Students analyze optimization problems, constraints, objective functions, and decision variables to identify effective mathematical solutions for complex real-life situations.
		PO3	Students design structured mathematical models and apply simplex, transportation, assignment, and game theory methods to develop optimized computational and managerial solutions.
		PO4	Students investigate alternative optimization strategies, compare feasible solutions, interpret outcomes, and evaluate efficiency using systematic mathematical approaches.
		PO5	Students use optimization software, spreadsheets, graphing tools, and computational applications for solving linear programming and Operations Research problems efficiently.
		PO6	Optimization and decision-making techniques support effective resource utilization, cost minimization, and ethical management practices in industrial and societal applications.
		PSO1	Operations Research techniques are implemented using computational tools, programming methods, and analytical software widely used in business intelligence and software systems.
		PSO2	Optimization techniques are extensively applied in logistics, project management, artificial intelligence, scheduling systems, and modern technological decision-support systems.
CO4	Analyze the structure and properties of lattices and Boolean algebras, and apply these algebraic concepts to understand logical systems, digital circuits, and their applications in computer science.	PO1	Lattices and Boolean algebra form the mathematical basis of digital logic, switching theory, computer architecture, and logical computation in computer science.
		PO2	Students analyze logical structures, algebraic properties, and Boolean relationships to solve problems related to logic simplification and computational systems.
		PO3	Students apply Boolean algebra and lattice concepts in the design and simplification of logical circuits and computational models used in digital systems.
		PO4	Students investigate algebraic properties, verify logical equivalences, and analyze truth tables and logical systems using mathematical reasoning.

		PO5	Students use simulation tools, logical circuit software, and visualization platforms for understanding Boolean operations and digital circuit representations.
		PO6	Logical reasoning and accurate computational design contribute to reliable and responsible development of digital and computing systems.
		PSO1	Boolean algebra and logical operations are directly used in programming constructs, digital systems, computer organization, and software logic development.
		PSO2	Students apply logical structures in designing innovative digital systems, computational models, and technology-based problem-solving applications.

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for Foundation of Mathematics (MDC5001C)		
SDG No and Title	Relevance to Course	Justification
SDG 4 – Quality Education	Directly relevant — promotes mathematical literacy and higher-order thinking skills	Combinatorics, numerical methods, OR, and algebra provide rigorous training that contributes to equitable, quality STEM education accessible through open resources.
SDG 9 – Industry, Innovation and Infrastructure	Relevant — mathematical foundations underpin engineering and computing innovations	Numerical methods and operations research methods are core to solving engineering design, optimization, and infrastructure planning problems.
SDG 8 – Decent Work and Economic Growth	Moderately relevant — optimization techniques aid economic decision-making	Linear Programming and Game Theory support business optimization, resource allocation, and strategic decision-making critical for economic productivity.

Explanation:

Unit 1: Introduction to Combinatorics and Its Applications

This unit contributes significantly to SDG 4 by developing students' mathematical reasoning, logical thinking, and analytical problem-solving skills through topics such as counting principles, permutations, combinations, recurrence relations, generating functions, and combinatorial applications. Concepts like the Pigeonhole Principle, Inclusion-Exclusion Principle, and Binomial Theorem enhance students' ability to think systematically and solve computational problems efficiently. The unit promotes

mathematical literacy, critical thinking, and independent learning, which are essential for quality education and strong foundations in computer applications and data analysis.

Unit 2: Numerical Methods

This unit strongly aligns with SDG 9 by introducing computational techniques and numerical algorithms used in scientific computing, engineering analysis, and technological problem-solving. Topics such as error analysis, numerical solutions of algebraic and transcendental equations, iterative techniques, and interpolation methods help students understand approximation and computational accuracy in real-world applications. Methods including Newton-Raphson, Gauss-Seidel, and interpolation techniques are widely applied in engineering simulations, software development, infrastructure modeling, and research-based innovations. The unit strengthens computational efficiency, algorithmic thinking, and innovation-oriented problem-solving abilities.

Unit 3: Operations Research and Linear Programming

This unit supports SDG 8 by providing students with optimization and decision-making techniques essential for efficient industrial and economic operations. Topics such as Linear Programming Problems (LPP), Transportation Problems, Assignment Problems, Simplex Method, and Game Theory develop analytical and managerial skills required for resource optimization, production planning, logistics, and strategic decision-making. Real-life applications of Operations Research help students understand practical business and industrial challenges, thereby improving employability, productivity, and problem-solving competence required for sustainable economic growth and professional development.

Unit 4: Lattices and Boolean Algebra

This unit contributes to SDG 9 through its focus on mathematical structures used in digital systems, logic design, and computer science applications. Concepts such as partially ordered sets, lattices, Boolean algebra, and their properties form the theoretical foundation of digital electronics, switching circuits, and computational logic. Applications of Boolean Algebra in computer science and digital logic support innovation in computing technologies and intelligent systems. The unit enhances abstract reasoning, logical analysis, and technology-oriented learning necessary for infrastructure development and advancements in computer applications.

Overall, the course effectively supports SDGs 4, 8, and 9 by integrating mathematical theory, computational techniques, optimization methods, and logical structures that are essential for modern technological and industrial environments. The course develops analytical thinking, computational efficiency, innovation, and professional problem-solving skills through combinatorics, numerical methods, operations research, and Boolean algebra. These competencies enhance students' academic growth, employability, and readiness for technology-driven careers while supporting sustainable innovation and economic productivity. The course structure also aligns with the academic vision and mission of Silver Oak College of Computer Application by promoting computational excellence, analytical ability, and industry-oriented education.